



Grade 11 Geography

2. Horizontal Rock Strata

Learning Guide



1. Key Concepts You Should Know

Weathering

- **Weathering** is the **breakdown of rocks into smaller pieces**.
- It happens while the rock is still in the same place.

Types of Weathering

- **Mechanical weathering**: Rocks break physically, for example because of temperature changes.
- **Chemical weathering**: Rocks change chemically, for example due to acid rain.
- **Biological weathering**: Living things break rocks, for example plant roots growing into cracks.

Erosion

- **Erosion** is the **removal of rock material** by agents such as: Wind, Water & Ice

Topography

- **Topography** means the **shape of the land**, also called **relief**.

Horizontally Layered Rocks

- These are **sedimentary rock layers** that were formed on **sea or flood plains**.
- They are arranged in flat layers, one on top of another.



2. What Are Horizontal Rock Strata?

- **Horizontal rock strata** are made of **sedimentary layers deposited over time**.
- Each layer represents material that settled and hardened at different times.
- The layers are separated by **bedding planes**.

Bedding Planes

- **Bedding planes** are lines or surfaces that separate one rock layer from another.
- They show that the rock was formed in layers over time.

Different Resistance to Erosion

Not all layers erode at the same rate:

- **Sandstone**
 - More resistant to erosion.
 - Forms hard, protective **caps**.
 - Helps create flat-topped landforms.
- **Shale**
 - Less resistant to erosion.
 - Erodes more easily.
 - Forms **gentler slopes** below the harder rock.



Key Idea:

Hard rock protects the top, while soft rock erodes faster underneath.



3. Topography of Horizontally Layered Rocks

The shape of the land depends on the **resistance of the rock layers**.

Steep Slopes

- Form where rocks are **resistant**.
- Resistant rocks do not erode easily, so they remain standing as steep cliffs or caps.

Gentle Slopes

- Form where rocks **erode easily**.
- Softer rocks are removed faster, creating smoother and gentler slopes.

Climate Influence

Climate also affects the appearance of the slopes:

- **Humid areas**
 - More moisture.
 - Slopes become **gentler**.
- **Arid areas**
 - Drier conditions.
 - Slopes become **steeper and more rugged**.



4. Flat-Topped Hill Formation

Flat-topped hills form when **hard and soft rock layers erode at different rates**.

Process

1. Horizontal sedimentary layers are exposed to weathering and erosion.
2. The **soft shale layers** erode faster.
3. The **hard sandstone layer** remains as a resistant cap.
4. The landform develops a **flat top** and steep sides.

Result

- A hill with a **flat upper surface**.
- Steeper sides where erosion has cut into the softer layers.



Important:

Flat-topped hills are common where resistant rock lies above weaker rock.



5. Basaltic Plateaux and Lava Flow Formation

- Basaltic plateaux are formed by **runny lava flows** spreading over large areas.
- When the lava cools, it hardens and forms **flat-topped landforms**.
- Basalt is very **resistant to erosion**.
- Only **powerful rivers** can cut through it.

Result

- Large, flat elevated areas.
- Rivers may cut deep valleys or **canyons** into the plateau.

Example

- **Drakensberg in South Africa**



Key Idea:

Lava spreads out, cools, hardens, and forms a resistant flat-topped plateau.



6. Karoo Landforms – Overview

Karoo landforms develop from **igneous intrusions**, especially **sills**.

How They Form

- Magma intrudes between **sedimentary rock layers**.
- It cools underground and forms **dolerite sills**.
- Dolerite is more resistant than the surrounding softer sedimentary rocks.
- Over time, erosion removes the softer rocks.
- The harder dolerite remains exposed as a **flat cap**.

Result

- Flat-topped hills on flat plains.
- Example shown in the document: **Platberg**.



Key Idea:

Dolerite acts like a protective cap, while softer rocks are removed by erosion.

7. Types of Karoo Landforms

Mesa

- A **wide, flat-topped hill**.
- The top is **wider than the height**.
- It is usually the larger landform in the sequence.

Butte

- A **smaller flat-topped hill**.
- The top is **narrower than the height**.
- It forms after a mesa has been further eroded.

Conical Hill

- A **rounded hill**.
- Forms from a **collapsed or heavily eroded butte**.
- It no longer has a clear flat top.

Formation Sequence

Plateau → Mesa → Butte → Conical Hill

8. Canyon vs Karoo Landscapes

Feature	Canyon	Karoo
Main Agent	Rivers through vertical erosion	Weathering and erosion
Dominant Rock	Horizontal layers	Dolerite sills and sedimentary rock
Common Landforms	Deep valleys	Mesa, butte, conical hill
Region	Arid	Semi-arid

9. Final Exam Summary

- **Horizontal rock strata** are sedimentary layers deposited over time.
- **Bedding planes** separate the layers.
- **Sandstone** is resistant and forms caps.
- **Shale** is less resistant and forms gentler slopes.
- Resistant rocks create **steep slopes**.
- Soft rocks create **gentle slopes**.
- **Humid climates** form gentler slopes.
- **Arid climates** form steep, rugged slopes.
- **Basaltic plateaux** form from lava flows.
- **Karoo landforms** develop from dolerite sills and erosion.
- Karoo landforms develop in this sequence:
Plateau → Mesa → Butte → Conical Hill

○

For the full interactive presentation, visit May Learning Hub:
www.maylearninghub.co.za